

The Kalman gain is obtained from Equation (8.37) as

$$\begin{aligned} K &= PC^T R^{-1} \\ &= P/R \\ \lim_{t \rightarrow \infty} K &= \sqrt{Q/R} \end{aligned} \quad (8.48)$$

The state estimate update expression is obtained from Equation (8.37) as

$$\begin{aligned} \dot{\hat{x}} &= A\hat{x} + Bu + K(y - C\hat{x}) \\ &= K(y - \hat{x}) \end{aligned} \quad (8.49)$$

From these expressions we see that if process noise increases (i.e., Q increases) then K increases. This is intuitively agreeable, because from the $\hat{x}$ equation we see that K defines the rate at which we change $\hat{x}$ based on the measurements. If Q is large then we have less confidence in our system model, and relatively more confidence in our measurements, so we change $\hat{x}$ more aggressively to be consistent with our measurements.

Similarly, we see that if we have large measurement noise (i.e., R is large) then K decreases. This is again intuitively agreeable. Large measurement noise means that we have less confidence in our measurements, so we change $\hat{x}$ less aggressively to be consistent with our measurements.

Finally, we see that P increases as both Q and R increase. An increase in the noise in either the system model or the measurements will degrade our confidence in our state estimate.

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8.3 ALTERNATE SOLUTIONS TO THE RICCATI EQUATION

The differential Riccati equation of Equation (8.37) can be computationally expensive to integrate, especially for systems with small time constants. Also, direct integration of the Riccati equation may result in a P matrix that loses its positive definiteness due to numerical problems. In this section we will look at some alternate solutions to the differential Riccati equation. This first two methods, called the transition matrix approach and the Chandrasekhar algorithm, are both intended to reduce computational effort. The third method, called square root filtering, is intended to reduce numerical difficulties.

8.3.1 The transition matrix approach

Assume that $P = \Lambda Y^{-1}$, where Λ and Y are $n \times n$ matrices to be determined. In the following we will determine what equalities must be satisfied by Λ and Y in order for this factorization to be valid. If the factorization is valid then

$$\begin{aligned} \dot{P} &= \dot{\Lambda} Y^{-1} + \Lambda \frac{d}{dt}(Y^{-1}) \\ &= \dot{\Lambda} Y^{-1} - \Lambda Y^{-1} \dot{Y} Y^{-1} \end{aligned} \quad (8.50)$$

where we have used Equation (1.51) for the time derivative of Y^{-1} . We post-multiply both sides of the above equation by Y to obtain

$$\dot{P}Y = \dot{\Lambda} - \Lambda Y^{-1} \dot{Y} \quad (8.51)$$

Recall from Equation (8.37) that the differential equation for P is given by

$$\dot{P} = AP + PA^T - PC^T R^{-1} CP + Q \quad (8.52)$$

Substitute ΛY^{-1} for P in this equation to obtain

$$\dot{P} = A\Lambda Y^{-1} + \Lambda Y^{-1} A^T - \Lambda Y^{-1} C^T R^{-1} C \Lambda Y^{-1} + Q \quad (8.53)$$

Post-multiply both sides of this equation by Y to obtain

$$\dot{P}Y = A\Lambda + \Lambda Y^{-1} A^T Y - \Lambda Y^{-1} C^T R^{-1} C \Lambda + QY \quad (8.54)$$

Now we can equate the right sides of Equations (8.51) and (8.54) to obtain

$$\begin{aligned} \dot{\Lambda} - \Lambda Y^{-1} \dot{Y} &= A\Lambda + \Lambda Y^{-1} A^T Y - \Lambda Y^{-1} C^T R^{-1} C \Lambda + QY \\ \dot{\Lambda} &= A\Lambda + QY + \Lambda Y^{-1} (\dot{Y} + A^T Y - C^T R^{-1} C \Lambda) \end{aligned} \quad (8.55)$$

This equation came from our original factorization of P , and if this equation reduces to $0 = 0$ then we know that the original factorization was valid. So if $\dot{Y} = C^T R^{-1} C \Lambda - A^T Y$, and $\dot{\Lambda} = A\Lambda + QY$, then our assumed factorization will be valid. These differential equations for Y and Λ can be combined as

$$\begin{aligned} \begin{bmatrix} \dot{\Lambda} \\ \dot{Y} \end{bmatrix} &= \begin{bmatrix} A & Q \\ C^T R^{-1} C & -A^T \end{bmatrix} \begin{bmatrix} \Lambda \\ Y \end{bmatrix} \\ &= J \begin{bmatrix} \Lambda \\ Y \end{bmatrix} \end{aligned} \quad (8.56)$$

where J is defined by the above equation. The initial conditions on Λ and Y can be chosen to be consistent with the initial condition on P as follows:

$$\begin{aligned} \Lambda(0) &= P(0) \\ Y(0) &= I \end{aligned} \quad (8.57)$$

Now suppose that A , Q , C , and R are constant (that is, we have an LTI system with constant process and measurement noise covariances). In this case J is constant and Equation (8.56) can be solved as

$$\begin{bmatrix} \Lambda(t+T) \\ Y(t+T) \end{bmatrix} = \exp(JT) \begin{bmatrix} \Lambda(t) \\ Y(t) \end{bmatrix} \quad (8.58)$$

This can be written as

$$\begin{bmatrix} \Lambda(t+T) \\ Y(t+T) \end{bmatrix} = \begin{bmatrix} \phi_{11}(T) & \phi_{12}(T) \\ \phi_{21}(T) & \phi_{22}(T) \end{bmatrix} \begin{bmatrix} \Lambda(t) \\ Y(t) \end{bmatrix} \quad (8.59)$$

where the ϕ_{ij} matrices are defined as the four $n \times n$ submatrices in $\exp(JT)$. From our original factorization assumption we have $\Lambda = PY$, so this equation can be written as

$$\begin{bmatrix} \Lambda(t+T) \\ Y(t+T) \end{bmatrix} = \begin{bmatrix} \phi_{11}(T) & \phi_{12}(T) \\ \phi_{21}(T) & \phi_{22}(T) \end{bmatrix} \begin{bmatrix} P(t)Y(t) \\ Y(t) \end{bmatrix} \quad (8.60)$$

This can be written as two separate equations:

$$\begin{aligned} \Lambda(t+T) &= \phi_{11}(T)P(t)Y(t) + \phi_{12}(T)Y(t) \\ Y(t+T) &= \phi_{21}(T)P(t)Y(t) + \phi_{22}(T)Y(t) \end{aligned} \quad (8.61)$$

Since $\Lambda(t+T) = P(t+T)Y(t+T)$, we can write the first equation as

$$P(t+T)Y(t+T) = \phi_{11}(T)P(t)Y(t) + \phi_{12}(T)Y(t) \quad (8.62)$$

Substituting for $Y(t+T)$ from Equation (8.61) in the above equation gives

$$\begin{aligned} P(t+T) [\phi_{21}(T)P(t)Y(t) + \phi_{22}(T)Y(t)] &= \phi_{11}(T)P(t)Y(t) + \phi_{12}(T)Y(t) \\ P(t+T) [\phi_{21}(T)P(t) + \phi_{22}(T)] &= \phi_{11}(T)P(t) + \phi_{12}(T) \end{aligned} \quad (8.63)$$

This equation is finally solved for $P(t+T)$ as

$$P(t+T) = [\phi_{11}(T)P(t) + \phi_{12}(T)] [\phi_{21}(T)P(t) + \phi_{22}(T)]^{-1} \quad (8.64)$$

This may be a faster way to solve for P instead of integrating the Riccati equation. Note that we do not have to worry about the integration step size with this method. This method can be used to propagate from $P(t)$ to $P(t+T)$ in a single equation, for any values t and T .

■ EXAMPLE 8.3

Suppose that we want to estimate a gyroscope drift rate ϵ (assumed to be constant) given measurements of the gyro angle θ . The system and measurement model can be written as

$$\begin{aligned} \dot{\theta} &= \epsilon \\ y &= \theta + v \\ \begin{bmatrix} \dot{\theta} \\ \dot{\epsilon} \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \epsilon \end{bmatrix} \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \epsilon \end{bmatrix} + v \\ v &\sim (0, R) \end{aligned} \quad (8.65)$$

Direct use of the differential Riccati equation from Equation (8.37) gives

$$\begin{aligned} \dot{P} &= AP + PA^T - PC^T R^{-1} CP + Q \\ \begin{bmatrix} \dot{P}_{11} & \dot{P}_{12} \\ \dot{P}_{12} & \dot{P}_{22} \end{bmatrix} &= \begin{bmatrix} 2P_{12} - P_{11}^2/R & P_{22} - P_{11}P_{12}/R \\ P_{22} - P_{11}P_{12}/R & -P_{12}^2/R \end{bmatrix} \end{aligned} \quad (8.66)$$

We can solve for P by performing three numerical integrations (recall that P is symmetric). However, it would be difficult to find a closed-form solution for $P(t)$ from these coupled differential equations. A transition matrix approach to this problem would proceed as follows, assuming that $P(0)$ is diagonal. We suppose that P is factored as $P = \Lambda Y^{-1}$, where Λ and Y are 2×2 matrices. The initial conditions on $\Lambda(t)$ and $Y(t)$ can be chosen as

$$\begin{aligned} \Lambda(0) &= P(0) \\ &= \begin{bmatrix} P_{11}(0) & 0 \\ 0 & P_{22}(0) \end{bmatrix} \\ Y(0) &= I \end{aligned} \quad (8.67)$$

The differential equation for $\Lambda(t)$ and $Y(t)$ is given as

$$\begin{bmatrix} \dot{\Lambda} \\ \dot{Y} \end{bmatrix} = J \begin{bmatrix} \Lambda \\ Y \end{bmatrix} \quad (8.68)$$

where the matrix J is computed as

$$\begin{aligned} J &= \begin{bmatrix} A & Q \\ C^T R^{-1} C & -A^T \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1/R & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \end{aligned} \quad (8.69)$$

The transition matrix for the differential equation for Λ and Y is computed as

$$\begin{aligned} \exp(Jt) &= \begin{bmatrix} 1 & t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ t/R & t^2/2R & 1 & 0 \\ -t^2/2R & -t^3/6R & -t & 1 \end{bmatrix} \\ &= \begin{bmatrix} \phi_{11}(t) & \phi_{12}(t) \\ \phi_{21}(t) & \phi_{22}(t) \end{bmatrix} \end{aligned} \quad (8.70)$$

where the $\phi_{ij}(t)$ terms are 2×2 matrix partitions. The Riccati equation solution is obtained from Equation (8.64) as

$$\begin{aligned} P(t) &= [\phi_{11}(t)P(0) + \phi_{12}(t)][\phi_{21}(t)P(0) + \phi_{22}(t)]^{-1} \\ &= \begin{bmatrix} P_{11}(0) & tP_{22}(0) \\ 0 & P_{22}(0) \end{bmatrix} \frac{1}{\Delta} \begin{bmatrix} 12R^2 - 2t^3P_{22}(0) & -6Rt^2P_{22}(0) \\ 12R^2t + 6t^2P_{11}(0) & 12R^2 + 12tP_{11}(0) \end{bmatrix} \end{aligned} \quad (8.71)$$

where Δ is given as

$$\Delta = 12R^2 + P_{11}(0)P_{22}(0)t^4 + 12P_{11}(0)tR + 4P_{22}(0)t^3R \quad (8.72)$$

Carrying out the multiplication and some algebra gives the Riccati equation solution as

$$\begin{aligned} P(t) &= \begin{bmatrix} P_{11}(t) & P_{12}(t) \\ P_{12}(t) & P_{22}(t) \end{bmatrix} \\ P_{11}(t) &= \frac{1}{\Delta} 4R [P_{11}(0)P_{22}(0)t^3 + 3P_{11}(0)R + 3t^2P_{22}(0)R] \\ P_{12}(t) &= \frac{1}{\Delta} 6RP_{22}(0)t [P_{11}(0)t + 2R] \\ P_{22}(t) &= \frac{1}{\Delta} 12RP_{22}(0) [P_{11}(0)t + R] \end{aligned} \quad (8.73)$$

With the transition matrix approach we have obtained a closed-form solution for $P(t)$, something that was not possible with a direct approach to the Riccati equation. In the special case that our initial uncertainty is infinite, we can

further simplify $P(t)$ as

$$\begin{aligned}\lim_{P(0) \rightarrow \infty} \Delta &= P_{11}(0)P_{22}(0)t^4 \\ \lim_{P(0) \rightarrow \infty} P(t) &= \begin{bmatrix} 4R/t & 6R/t^2 \\ 6R/t^2 & 12R/t^3 \end{bmatrix} \\ \lim_{t \rightarrow \infty} \left[\lim_{P(0) \rightarrow \infty} P(t) \right] &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\end{aligned}\quad (8.74)$$

That is, our uncertainty goes to zero as time goes to infinity. This occurs because the process noise is zero (i.e., we are estimating a constant). Since $K = PC^T R^{-1}$, we see that the Kalman gain also goes to zero as time goes to infinity. This simply means that eventually we get so many measurements that our knowledge is complete. Additional measurements cannot give us any new information, so we ignore additional measurements.

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8.3.2 The Chandrasekhar algorithm

Recall the differential Riccati equation for the continuous-time Kalman filter from Equation (8.37):

$$\dot{P} = AP + PA^T - PC^T R^{-1} CP + Q \quad (8.75)$$

If P were not symmetric then the numerical computation of P would require n^2 integrations. However, since $P = P^T$ the computation of P requires only $n(n+1)/2$ integrations. This can still be computationally taxing, especially for problems with small time constants. The Chandrasekhar algorithm gives computational savings in some circumstances. The algorithm is based on the work of the Nobel prize winning astrophysicist Subramanian Chandrasekhar, who used similar algorithms to solve computationally difficult astrophysics problems in the 1940s [Cha47, Cha48]. Chandrasekhar's algorithms were applied to Kalman filtering in [Kai73, Kai00]. The Chandrasekhar algorithm applies only when A , C , R , and Q are constant.

8.3.2.1 The Chandrasekhar algorithm derivation Consider the continuous-time differential equation for the state estimate, assuming that the original system is time-invariant and the Kalman gain K is a constant:

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + K(y - C\hat{x}) \\ &= (A - KC)\hat{x} + Ky\end{aligned}\quad (8.76)$$

The measurement y is the output of the system, but it is the input to the filter. Consider the zero-input Kalman filter (i.e., $y = 0$).

$$\dot{\hat{x}} = (A - KC)\hat{x} \quad (8.77)$$

This equation has the solution

$$\begin{aligned}\hat{x}(t) &= \exp[(A - KC)t]\hat{x}(0) \\ &= \phi(t)\hat{x}(0)\end{aligned}\quad (8.78)$$

where $\phi(t)$ is the state transition matrix of the filter and is defined by the above equation. From the definition of $\phi(t)$ as a state transition matrix we know that

$$\begin{aligned}\dot{\phi} &= (A - KC)\phi \\ \phi(0) &= I\end{aligned}\tag{8.79}$$

We can differentiate both sides of Equation (8.75) to obtain

$$\begin{aligned}\ddot{P} &= A\dot{P} + \dot{P}A^T - \dot{P}C^TR^{-1}CP - PC^TR^{-1}C\dot{P} \\ &= A\dot{P} + \dot{P}A^T - \dot{P}C^TK^T - KC\dot{P} \\ &= (A - KC)\dot{P} + \dot{P}(A - KC)^T\end{aligned}\tag{8.80}$$

Now note that for a general time-varying matrix $Y(t)$, if $\dot{Y} = AY + YA^T$, where A is a constant matrix, then $Y(t) = \exp(At)Y(0)\exp(A^Tt)$ (see Problem 8.2). Therefore, we can solve the above equation for $\dot{P}$ as

$$\dot{P} = \phi\dot{P}(0)\phi^T\tag{8.81}$$

where $\dot{P}(0)$ is obtained from Equation (8.75) as

$$\dot{P}(0) = AP(0) + P(0)A^T - P(0)C^TR^{-1}CP(0) + Q\tag{8.82}$$

The symmetric matrix $\dot{P}(0)$ can be factored as follows (see Section 8.3.2.2):

$$\dot{P}(0) = M_1M_1^T - M_2M_2^T\tag{8.83}$$

$\dot{P}(0)$ is an $n \times n$ matrix. The rank of $\dot{P}(0)$ is $\alpha \leq n$. Since $\dot{P}(0)$ is symmetric, all of its eigenvalues are real. The number of positive eigenvalues of $\dot{P}(0)$ is β , and the number of negative eigenvalues is $(\alpha - \beta)$. Matrix M_1 is an $n \times \beta$ matrix, and M_2 is an $n \times (\alpha - \beta)$ matrix. From the previous three equations we can write

$$\begin{aligned}\dot{P} &= \phi\dot{P}(0)\phi^T \\ &= \phi(M_1M_1^T - M_2M_2^T)\phi^T \\ &= \phi M_1M_1^T\phi^T - \phi M_2M_2^T\phi^T\end{aligned}\tag{8.84}$$

Now define the matrices Y_1 and Y_2 as

$$\begin{aligned}Y_1 &= \phi M_1 \\ Y_2 &= \phi M_2\end{aligned}\tag{8.85}$$

Then the $\dot{P}$ equation can be written as

$$\dot{P} = Y_1Y_1^T - Y_2Y_2^T\tag{8.86}$$

Also, from the definition of Y_1 we can see that

$$\begin{aligned}Y_1(0) &= \phi(0)M_1 = M_1 \\ \dot{Y}_1 &= \dot{\phi}M_1 \\ &= (A - KC)\phi M_1 \\ &= (A - KC)Y_1\end{aligned}\tag{8.87}$$

Similarly, we see that

$$\begin{aligned} Y_2(0) &= \phi(0)M_2 = M_2 \\ \dot{Y}_2 &= (A - KC)Y_2 \end{aligned} \quad (8.88)$$

Recall from Equation (8.37) that $K = PC^TR^{-1}$. Therefore, a differential equation and initial condition for K can be written as

$$\begin{aligned} \dot{K} &= \dot{P}C^TR^{-1} \\ &= (Y_1Y_1^T - Y_2Y_2^T)C^TR^{-1} \\ K(0) &= P(0)C^TR^{-1} \end{aligned} \quad (8.89)$$

To compute K from its differential equation we need to integrate three equations.

1. We need to integrate Y_1 from Equation (8.87), where Y_1 is an $n \times \beta$ matrix.
2. We need to integrate Y_2 from Equation (8.88), where Y_2 is an $n \times (\alpha - \beta)$ matrix.
3. We need to integrate K from Equation (8.89), where K is an $n \times r$ matrix (r is the number of measurements of the system).

So we need to perform a total of $n(\alpha + r)$ integrations. The direct computation of P from the differential Riccati equation requires $n(n + 1)/2$ integrations. So if $2(\alpha + r) < (n + 1)$ then the Chandrasekhar algorithm reduces the computational effort of solving the differential Riccati equation.

The Chandrasekhar algorithm

The Chandrasekhar algorithm can be summarized as follows.

1. Compute $\dot{P}(0)$.
2. Use the method of Section 8.3.2.2 to find M_1 and M_2 matrices that satisfy $\dot{P}(0) = M_1M_1^T - M_2M_2^T$.
3. Initialize $Y_1(0) = M_1$, $Y_2(0) = M_2$, and $K(0) = P(0)C^TR^{-1}$.
4. Integrate K , Y_1 , and Y_2 as follows:

$$\begin{aligned} \dot{K} &= (Y_1Y_1^T - Y_2Y_2^T)C^TR^{-1} \\ \dot{Y}_1 &= (A - KC)Y_1 \\ \dot{Y}_2 &= (A - KC)Y_2 \end{aligned} \quad (8.90)$$

8.3.2.2 Chandrasekhar factorization The derivation of the Chandrasekhar algorithm requires the factorization of $\dot{P}(0)$ as shown in Equation (8.83):

$$\dot{P}(0) = M_1M_1^T - M_2M_2^T \quad (8.91)$$

$\dot{P}(0)$ is an $n \times n$ matrix with rank $\alpha \leq n$. The number of positive eigenvalues of $\dot{P}(0)$ is β , and the number of negative eigenvalues is $(\alpha - \beta)$. Matrix M_1 is an $n \times \beta$

matrix, and M_2 is an $n \times (\alpha - \beta)$ matrix. In this section, we will show one way to perform that factorization.

Since $\dot{P}(0)$ is symmetric, all of its eigenvalues are real. We can therefore write the Jordan form of $\dot{P}(0)$ as

$$\begin{aligned}\dot{P}(0) &= SDS^T \\ &= \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \begin{bmatrix} D_1 & 0 & 0 \\ 0 & -D_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} S_{11}^T & S_{21}^T & S_{31}^T \\ S_{12}^T & S_{22}^T & S_{32}^T \\ S_{13}^T & S_{23}^T & S_{33}^T \end{bmatrix} \quad (8.92)\end{aligned}$$

S is an orthogonal matrix whose columns comprise the eigenvectors of $\dot{P}(0)$. The $\beta \times \beta$ matrix D_1 is a diagonal matrix whose entries are the positive eigenvalues of $\dot{P}(0)$. The $(\alpha - \beta) \times (\alpha - \beta)$ matrix D_2 is a diagonal matrix whose entries are the magnitudes of the negative eigenvalues of $\dot{P}(0)$. Multiplying out the above equation results in

$$\dot{P}(0) = N_1 + N_2 \quad (8.93)$$

where N_1 and N_2 are given as

$$\begin{aligned}N_1 &= \begin{bmatrix} S_{11}D_1S_{11}^T & S_{11}D_1S_{21}^T & S_{11}D_1S_{31}^T \\ S_{21}D_1S_{11}^T & S_{21}D_1S_{21}^T & S_{21}D_1S_{31}^T \\ S_{31}D_1S_{11}^T & S_{31}D_1S_{21}^T & S_{31}D_1S_{31}^T \end{bmatrix} \\ &= \begin{bmatrix} S_{11} \\ S_{21} \\ S_{31} \end{bmatrix} D_1 \begin{bmatrix} S_{11}^T \\ S_{21}^T \\ S_{31}^T \end{bmatrix} \\ N_2 &= \begin{bmatrix} S_{12}D_2S_{12}^T & S_{12}D_2S_{22}^T & S_{12}D_2S_{32}^T \\ S_{22}D_2S_{12}^T & S_{22}D_2S_{22}^T & S_{22}D_2S_{32}^T \\ S_{32}D_2S_{12}^T & S_{32}D_2S_{22}^T & S_{32}D_2S_{32}^T \end{bmatrix} \\ &= \begin{bmatrix} S_{12} \\ S_{22} \\ S_{32} \end{bmatrix} D_2 \begin{bmatrix} S_{12}^T \\ S_{22}^T \\ S_{32}^T \end{bmatrix} \quad (8.94)\end{aligned}$$

Note that N_1 is the product of an $n \times \beta$ matrix, the $\beta \times \beta$ matrix D_1 , and a $\beta \times n$ matrix. N_1 can therefore be written as

$$N_1 = M_1 M_1^T \quad (8.95)$$

where M_1 is the $n \times \beta$ matrix

$$M_1 = \begin{bmatrix} S_{11} \\ S_{21} \\ S_{31} \end{bmatrix} \sqrt{D_1} \quad (8.96)$$

A similar development can be followed to see that M_2 is the $n \times (\alpha - \beta)$ matrix

$$M_2 = \begin{bmatrix} S_{12} \\ S_{22} \\ S_{32} \end{bmatrix} \sqrt{D_2} \quad (8.97)$$

8.3.3 The square root filter

The early days of Kalman filtering in the 1960s saw a lot of successful applications. But there were also some problems in implementation, many due to numerical difficulties. The differential Riccati equation solution $P(t)$ should theoretically always be a symmetric positive semidefinite matrix (since it is a covariance matrix). But numerical problems in computer implementations sometimes led to P matrices that became indefinite or nonsymmetric. This was often because of the short word lengths in the computers of the 1960s [Sch81].¹ This led to a lot of research during that decade related to numerical implementations.

Now that computers have become so much more capable, we don't have to worry about numerical problems as often. Nevertheless, numerical issues still arise in finite word-length implementations of algorithms, especially in embedded systems.² The square root filter was developed in order to effectively increase the numerical precision of the Kalman filter and hence mitigate numerical difficulties in implementations.

The square root filter is based on the idea of finding an S matrix such that $P = SS^T$. The S matrix is then called a square root of P . Note that the definition of the square root of P is *not* that $P = S^2$, but rather $P = SS^T$. Also note that this definition of the matrix square root is not standard. Some books and papers define the matrix square root as $P = S^2$, others define it as $P = S^T S$, and others define it as $P = SS^T$. The latter definition is the one that we will use in this book. Finally, note that the square root of a matrix may not be unique; that is, there may be more than one solution for S in the equation $P = SS^T$. (This is analogous to the existence of multiple square roots for scalars. For example, the number 4 has two square roots: +2 and -2.) Sections 6.3 and 6.4 contain a discussion of square root filtering for the discrete-time Kalman filter.

After defining S as the square root of P , we will integrate S instead of P in our Kalman filter solution. This requires more computational effort but it doubles the precision of the filter and helps prevent numerical problems. From the differential Riccati equation of Equation (8.37), and the definition of S , we obtain

$$\begin{aligned}\dot{P} &= AP + PA^T - PC^T R^{-1} CP + Q \\ \dot{S}S^T + S\dot{S}^T &= ASS^T + SS^T A^T - SS^T C^T R^{-1} CSS^T + Q\end{aligned}\quad (8.98)$$

Now premultiply both sides by S^{-1} and postmultiply by S^{-T} to obtain

$$\begin{aligned}S^{-1}\dot{P}S^{-T} &= S^{-1}\dot{S} + \dot{S}^T S^{-T} \\ &= S^{-1}AS + S^T A^T S^{-T} - S^T C^T R^{-1} CS + S^{-1}QS^{-T}\end{aligned}\quad (8.99)$$

Since P is symmetric positive definite, we can always find an upper triangular S such that $P = SS^T$ [Gol89, Moo00]. For example, consider the following matrices:

¹The United States' Apollo space program of the 1960s resulted in the first man on the moon in 1969. The Apollo spacecraft guidance computer had a word length of 16 bits [Bat82], which corresponds to 4.8 decimal digits of precision.

²Most microcontrollers in the first decade of the 21st century have 16 bit words, and 8 bit microcontrollers still comprise a large share of the market.